REPORT 012: NKEDITOR4 — EPOCH 012: MULTI-PAGE & MODIFIER LAYER ENGINE WITH SOFT TAKEOVER

2026-11-04 01:13:53 | 4-Page Controller Layer Engine (src/layer_manager.rs), Cycle+Track Modifier Combos, Automatic Cross-Page Soft Takeover, and /nkeditor/layer/select OSC Endpoint

Operator: Antigravity Agent · Follows: Report 011

Prompt

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Implement a Multi-Page & Modifier Layer Engine in `src/layer_manager.rs` so compact
↪  8-knob/8-slider controllers like the Korg nanoKONTROL2 can control deep multi-row
↪  parameter sets (surpassing the 3-knob-row Akai MIDImix): provide 4 switchable
↪  controller pages (`Page 1`: Voice Density & Level, `Page 2`: Timbre Morph &
↪  Decay, `Page 3`: Pitch/Spectra & Noise Filter, `Page 4`: Global Feel, Swing,
↪  Humanize & Master FX) selectable via nanoKONTROL2 Marker Left/Right (`CC 61/62`),
↪  Cycle+Track combos, GUI page tabs, or OSC (`/nkeditor/layer/select`).
↪  Automatically engage Soft Takeover (Catch/Pickup) and update hardware LEDs
↪  whenever the active page changes so physical knob positions never cause parameter
↪  jumps across pages. Add automated tests (with `CARGO_BUILD_JOBS=2`), git commit
↪  all changes, and compile `reports/report-012.html` from `reports/report-012.md`
↪  via `/home/renick/claude/html-report-generator/build-report.sh --pdf` and commit.
```

TL;DR

Implemented the Multi-Page & Modifier Layer Engine in src/layer_manager.rs, enabling compact 8-knob/8-slider controllers like the Korg nanoKONTROL2 to control 64 continuous parameters across 4 switchable pages (Page 1: Voice Density & Level, Page 2: Timbre Morph & Decay, Page 3: Pitch/Spectra & Noise Filter, Page 4: Global Feel, Swing, Humanize & Master FX). Supports switching via Marker Left/Right (CC 61/62), Cycle+Track (CC 46 + CC 58/59) and Cycle+Solo 1..4 combos, GUI page tabs, and OSC (/nkeditor/layer/select), automatically engaging Soft Takeover and syncing hardware LEDs on every page switch. All 45 Rust tests and 10 Node.js tests pass (55 total).

Plain-English Overview: What Changed and Why It Matters

An Akai `MIDI Mix` has 3 rows of 8 knobs plus 1 row of 8 faders (32 continuous controls), whereas a Korg `nanoKONTROL2` is much more compact and portable—offering 1 row of 8 knobs and 1 row of 8 sliders (16 continuous controls) plus a full 11-button transport deck. Without a paging system, a `nanoKONTROL2` performer can only control 16 continuous parameters at a time. Worse, if a naive paging system simply switches what a physical knob controls, turning a knob that was left at `110` on Page 1 will immediately cause a jarring jump if the parameter on Page 2 was sitting at `64`.

In Epoch 12, we built `src/layer_manager.rs` to give the Korg `nanoKONTROL2` **4 switchable controller pages (64 continuous controls total—twice as many as an Akai MIDImix)** with **automatic cross-page Soft Takeover protection**:

- **Page 1 — Voice Density & Level (`VoiceDensityAndLevel`)**: Knobs `1..8` control per-voice note density (`CC 16..23`, Ch 1); Sliders `1..8` control per-voice level/gain (`CC 0..7`, Ch 1).
- **Page 2 — Timbre Morph & Decay (`TimbreMorphAndDecay`)**: Knobs `1..8` control per-voice FM

timbre morph/index (`CC 24..31`, Ch 1); Sliders `1..8` control per-voice envelope decay (`CC 8..15`, Ch 1).

- **Page 3 — Pitch/Spectra & Noise Filter (`PitchSpectraAndNoiseFilter`)**: Knobs `1..8` control per-voice harmonic spectra (`CC 30` Nord Drum 2 `TONE_SPECTRA` on Channels `1..8`); Sliders `1..8` control per-voice noise filter cutoff (`CC 14` Nord Drum 2 `NOISE_FILTER_FREQ` on Channels `1..8`).
- **Page 4 — Global Feel, Swing, Humanize & Master FX (`GlobalFeelAndMasterFx`)**: Knobs `1..8` control Global Density, Feel Scale, Swing Depth, Humanize, Distortion Drive, EQ Gain, Delay Send, and Reverb Mix (`CC 58..65`, Ch 10); Sliders `1..8` control Master Bus Level, Sub Lift, Transient Punch, Stereo Width, Delay Feedback, Reverb Decay, Limiter Threshold, and Output Trim (`CC 70..77`, Ch 10).
- **Marker Left / Marker Right (`CC 61 / CC 62`) Page Stepping**: Single-button step backward or forward across Pages `1..4`.
- **Cycle + Track / Solo Modifier Combos**: Holding `Cycle` (`CC 46`) lights the Cycle LED and turns `Track Left/Right` (`CC 58/59`) into page steppers and `Solo 1..4` (`CC 32..35`) into direct jump buttons for Pages `1..4`.
- **Interactive `egui` Layer Page Selector Bar (`render_layer_page_selector_bar`)**: One-click GUI page tabs displaying the active page, `CYCLE MOD` status, Soft Takeover lock counter (`X/16 LOCKED`), and an 8-column parameter legend for all 8 knobs and 8 sliders.
- **Bidirectional UDP OSC (`/nkeditor/layer/select`)**: Added `OscCommand::SelectLayer` in `src/osc_server.rs` and `selectLayer(page)` in `js/nkeditor-sdk.js`.
- **Zero-Jump Soft Takeover Across Page Switches (`MultiPageLayerManager::select_page`)**: Whenever the user switches from Page A to Page B, `select_page` snapshots Page A's current targets, ensures Soft Takeover (`Pickup` or `Catch`) is active on `DeviceMappingPipeline`, loads Page B's stored targets against the current physical knob/slider positions, and locks any mismatched control until the physical knob crosses or catches Page B's target value.

Plain-English Layer Feature	Rust Symbol ('src/layer_manager.rs')	Hardware / OSC Trigger	Technical Behavior
4 Controller Pages	`ControllerPage` (`ALL: [Self; 4]`)	Pages `1..4` (64 continuous controls)	Models Density/Level, Morph/Decay, Pitch/Filter, and Global Feel/Master FX
Per-Page Parameter Bank	`PageParameterBank::for_page`	Stores `DeviceMappingConfig` + `continuous_targets`	Independent CC/Channel mappings, 16 continuous targets, and button states per page
Cross-Page Soft Takeover	`MultiPageLayerManager::select_page`	Automatic on every page switch	Snapshots outgoing page, loads incoming targets, and locks mismatched physical knobs
Hardware Shortcut Handler	`MultiPageLayerManager::handle_nanokontrol2_cc`	`CC 61/62`, `CC 46 + CC 58/59`, `CC 46 + CC 32..35`	Tracks `cycle_modifier_held`, switches pages, and illuminates `track1..4.solo` LED
`egui` Layer Selector Bar	`render_layer_page_selector_bar`	4 GUI tabs + 8-column K/S label strip	Renders crisp `Rounding::ZERO` tabs and live `X/16 LOCKED` takeover indicator

Plain-English Layer Feature	Rust Symbol (‘src/layer_manager.rs‘)	Hardware / OSC Trigger	Technical Behavior
OSC Layer Endpoint	`OscCommand::SelectLayer` / `NkEditorClient.selectLayer`	`/nkeditor/layer/select`	Switches or broadcasts active layer page over UDP OSC

Tests

QA

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**Cross-Page Soft Takeover Verification**: Confirmed in `test_four_controller_pages_and_soft_takeover_lock_on_page_switch`
that sweeping Knob 1 (`CC 16`) to `110` on Page 1 and switching to Page 2 (where target is `64` on `CC 24`)
preserves `110` in Page 1's bank, locks Knob 1 on Page 2 (`SoftTakeoverLocked` when moved to `100`), unlocks
and emits `CC 24 = 60` once swept across `64`, and re-locks against `110` when returning to Page 1.

**Modifier & LED Verification**: Confirmed that `Marker Right` (`CC 62`), `Cycle + Track Right` (`CC 46
+ CC 59`), and `Cycle + Solo 4` (`CC 46 + CC 35`) switch pages, update `track1.solo..track4.solo` and
`transport.cycle` LEDs, and broadcast `/nkeditor/layer/select` over UDP OSC.
```

Files touched

- ‘src/layer_manager.rs‘ (Created — 852 lines: ‘ControllerPage‘, ‘PageParameterBank‘, ‘LayerShortcutOutcome‘, ‘MultiPageLayerManager‘, ‘render_layer_page_selector_bar‘, and 3 unit/integration tests)
- ‘src/osc_server.rs‘ (Modified — added ‘OscCommand::SelectLayer‘ and ‘/nkeditor/layer/select‘ parser/handler)
- ‘src/lib.rs‘ (Modified — exported ‘pub mod layer_manager;‘)
- ‘js/nkeditor-sdk.js‘ (Modified — added ‘OSC_ENDPOINTS.layerSelect‘ and ‘NkEditorClient.prototype.selectLay

Test coverage (55 total)

QA verdict

PASS